

How to calculate floor sum?

 Правка en16, от [YipChip](#), 2025-03-24 15:24:05

The floor sum is an algorithm that solves for the value of a function approximating

$$f(a, b, c, n) = \sum_{i=0}^n \left\lfloor \frac{ai+b}{c} \right\rfloor \text{ in } O(\log n) \text{ time complexity.}$$

Maybe you haven't heard of it, but you can get the algorithm by simple mathematical derivation.

To calculate the sum of the functions, $n \leq 10^9$.

$$\begin{aligned} f(a, b, c, n) &= \sum_{i=0}^n \left\lfloor \frac{ai+b}{c} \right\rfloor \\ g(a, b, c, n) &= \sum_{i=0}^n \left\lfloor \frac{ai+b}{c} \right\rfloor^2 \\ h(a, b, c, n) &= \sum_{i=0}^n i \left\lfloor \frac{ai+b}{c} \right\rfloor \end{aligned}$$

(I) when $a < c$ and $b < c$.

First simplify $f(a, b, c, n)$.

$$\begin{aligned} f(a, b, c, n) &= \sum_{i=0}^n \left\lfloor \frac{ai+b}{c} \right\rfloor \\ &= \sum_{i=0}^n \sum_{j=1}^{\left\lfloor \frac{ai+b}{c} \right\rfloor} \left[j \leq \left\lfloor \frac{ai+b}{c} \right\rfloor \right] \\ &= \sum_{i=0}^n \sum_{j=1}^{\left\lfloor \frac{ai+b}{c} \right\rfloor} \left[j \leq \left\lfloor \frac{ai+b}{c} \right\rfloor \right] \\ &= \sum_{i=0}^n \sum_{j=0}^{\left\lfloor \frac{ai+b}{c} \right\rfloor - 1} \left[j+1 \leq \left\lfloor \frac{ai+b}{c} \right\rfloor \right] \\ &= \sum_{i=0}^n \sum_{j=0}^{\left\lfloor \frac{ai+b}{c} \right\rfloor - 1} \left[cj+c \leq ai+b \right] \\ &= \sum_{i=0}^n \sum_{j=0}^{\left\lfloor \frac{ai+b}{c} \right\rfloor - 1} \left[cj+c-b \leq ai \right] \\ &= \sum_{i=0}^n \sum_{j=0}^{\left\lfloor \frac{ai+b}{c} \right\rfloor - 1} \left[cj+c-b-1 < ai \right] \\ &= \sum_{i=0}^n \sum_{j=0}^{\left\lfloor \frac{ai+b}{c} \right\rfloor - 1} \left[\left\lfloor \frac{cj+c-b-1}{a} \right\rfloor < i \right] \\ &= \sum_{j=0}^{\left\lfloor \frac{an+b}{c} \right\rfloor - 1} \sum_{i=0}^n \left[\left\lfloor \frac{cj+c-b-1}{a} \right\rfloor < i \right] \\ &= \sum_{j=0}^{\left\lfloor \frac{an+b}{c} \right\rfloor - 1} \sum_{i=\left\lfloor \frac{cj+c-b-1}{a} \right\rfloor + 1}^n 1 \\ &= \sum_{j=0}^{\left\lfloor \frac{an+b}{c} \right\rfloor - 1} \left(n - \left\lfloor \frac{cj+c-b-1}{a} \right\rfloor \right) \\ &= n \left\lfloor \frac{an+b}{c} \right\rfloor - \sum_{j=0}^{\left\lfloor \frac{an+b}{c} \right\rfloor - 1} \left\lfloor \frac{cj+c-b-1}{a} \right\rfloor \\ &= n \left\lfloor \frac{an+b}{c} \right\rfloor - f(c, c+b-1, a, \left\lfloor \frac{an+b}{c} \right\rfloor - 1) \end{aligned}$$

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5	tourist	3530
6	Kevin114514	3510
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Let $m = \left\lfloor \frac{an+b}{c} \right\rfloor$, then we get:

$$f(a, b, c, n) = nm - f(c, c - b - 1, a, m - 1)$$

For the above derivation, we call it **Lemma 1**:

$$\text{Lemma 1: } \sum_{i=0}^n \left\lfloor \frac{ai+b}{c} \right\rfloor = \sum_{j=0}^{\left\lfloor \frac{an+b}{c} \right\rfloor - 1} \sum_{i=\left\lfloor \frac{cj+c-b-1}{a} \right\rfloor + 1}^n 1$$

Next consider the simplification of $h(a, b, c, n)$, which follows from **Lemma 1**:

$$\begin{aligned} h(a, b, c, n) &= \sum_{i=0}^n i \left\lfloor \frac{ai+b}{c} \right\rfloor \\ &= \sum_{j=0}^{\left\lfloor \frac{an+b}{c} \right\rfloor - 1} \sum_{i=\left\lfloor \frac{cj+c-b-1}{a} \right\rfloor + 1}^n i \end{aligned}$$

Let $t = \left\lfloor \frac{cj+c-b-1}{a} \right\rfloor$, $m = \left\lfloor \frac{an+b}{c} \right\rfloor$, then:

$$\begin{aligned} h(a, b, c, n) &= \sum_{j=0}^{\left\lfloor \frac{an+b}{c} \right\rfloor - 1} \sum_{i=\left\lfloor \frac{cj+c-b-1}{a} \right\rfloor + 1}^n i \\ &= \sum_{j=0}^{m-1} \sum_{i=t+1}^n i \\ &= \sum_{j=0}^{m-1} \frac{(t+n+1)(n-t)}{2} \end{aligned}$$

Lemma 1: Let $t = \left\lfloor \frac{cj+c-b-1}{a} \right\rfloor$, $m = \left\lfloor \frac{an+b}{c} \right\rfloor$, then:

$$\sum_{i=0}^n i \left\lfloor \frac{ai+b}{c} \right\rfloor = \sum_{j=0}^{m-1} \frac{(t+n+1)(n-t)}{2}$$

Since j in the expression for t is related to summation, we present t separately.

$$\begin{aligned} \frac{(t+n)(n-t+1)}{2} &= \frac{1}{2}(n^2 - t^2 - t + n) \\ &= \frac{1}{2}((n^2 + n) - (t^2 + t)) \end{aligned}$$

The original equation can then be simplified.

$$\begin{aligned} \sum_{j=0}^{m-1} \frac{(t+n+1)(n-t)}{2} &= \sum_{j=0}^{m-1} \frac{1}{2} \left((n^2 + n) - (t^2 + t) \right) \\ &= \frac{1}{2} \sum_{j=0}^{m-1} (n^2 + n) - \frac{1}{2} \sum_{j=0}^{m-1} (t^2 + t) \\ &= \frac{mn^2 + mn}{2} - \frac{1}{2} \sum_{j=0}^{m-1} t^2 - \frac{1}{2} \sum_{j=0}^{m-1} t \\ &= \frac{1}{2} \left(mn^2 + mn - \sum_{j=0}^{m-1} \left\lfloor \frac{cj+c-b-1}{a} \right\rfloor^2 - \sum_{j=0}^{m-1} \left\lfloor \frac{cj+c-b-1}{a} \right\rfloor \right) \\ &= \frac{1}{2} \left(mn(n+1) - g(c, c-b-1, a, m-1) - f(c, c-b-1, a, m-1) \right) \end{aligned}$$

Then consider the simplification of $g(a, b, c, n)$, where we know summation formula

$\sum_{i=1}^n i = \frac{n^2+n}{2}$, which can be inverted to get $2 \sum_{i=1}^n i - n = n^2$, whereupon our $g(a, b, c, n)$ can be turned into:

$$\begin{aligned} g(a, b, c, n) &= \sum_{i=0}^n \left\lfloor \frac{ai+b}{c} \right\rfloor^2 \\ &= \sum_{i=0}^n \left(2 \sum_{j=1}^{\left\lfloor \frac{ai+b}{c} \right\rfloor} j - \left\lfloor \frac{ai+b}{c} \right\rfloor \right) \\ &= \sum_{i=0}^n \left(2 \sum_{j=1}^{\left\lfloor \frac{ai+b}{c} \right\rfloor} j - f(a, b, c, n) \right) \end{aligned}$$

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According to **Lemma 2** it follows that for $t = \left\lfloor \frac{cj+c-b-1}{a} \right\rfloor$, $m = \left\lfloor \frac{an+b}{c} \right\rfloor$ and there are:

$$\begin{aligned}
 g(a, b, c, n) &= \sum_{i=0}^n \left(2 \sum_{j=1}^{\left\lfloor \frac{ai+b}{c} \right\rfloor} j - f(a, b, c, n) \right) \\
 &= 2 \sum_{i=0}^n \sum_{j=1}^{\left\lfloor \frac{ai+b}{c} \right\rfloor} j - \sum_{i=0}^n \left\lfloor \frac{ai+b}{c} \right\rfloor \\
 &= 2 \sum_{i=0}^n \sum_{j=0}^{\left\lfloor \frac{ai+b}{c} \right\rfloor - 1} (j+1) - f(a, b, c, n) \\
 &= 2 \sum_{i=0}^n \sum_{j=0}^{\left\lfloor \frac{ai+b}{c} \right\rfloor - 1} (j+1) \left[j \leq \left\lfloor \frac{ai+b}{c} \right\rfloor - 1 \right] - f(a, b, c, n) \\
 &= 2 \sum_{i=0}^n \sum_{j=0}^{m-1} (j+1) \left[j \leq \left\lfloor \frac{ai+b}{c} \right\rfloor - 1 \right] - f(a, b, c, n)
 \end{aligned}$$

From **Lemma 1**:

$$\begin{aligned}
 g(a, b, c, n) &= 2 \sum_{i=0}^n \sum_{j=0}^{m-1} (j+1) \left[\left\lfloor \frac{cj+c-b-1}{a} \right\rfloor < i \right] - f(a, b, c, n) \\
 &= 2 \sum_{j=0}^{m-1} (j+1) \left(n - \left\lfloor \frac{cj+c-b-1}{a} \right\rfloor \right) - f(a, b, c, n) \\
 &= 2 \sum_{j=0}^{m-1} (j+1)n - 2 \sum_{j=0}^{m-1} (j+1) \left\lfloor \frac{cj+c-b-1}{a} \right\rfloor - f(a, b, c, n) \\
 &= (m+1)mn - 2 \sum_{j=0}^{m-1} j \left\lfloor \frac{cj+c-b-1}{a} \right\rfloor - 2 \sum_{j=0}^{m-1} \left\lfloor \frac{cj+c-b-1}{a} \right\rfloor - f(a, b, c, n) \\
 &= (m+1)mn - 2 \left(h(c, c-b-1, a, m-1) - f(c, c-b-1, a, m-1) \right) - f(a, b, c, n)
 \end{aligned}$$

From this, when $a \leq c$ and $b \leq c$, we have:

$$\begin{aligned}
 f(a, b, c, n) &= nm - f(c, c-b-1, a, m-1) \\
 g(a, b, c, n) &= (m+1)mn - 2 \left(h(c, c-b-1, a, m-1) - f(c, c-b-1, a, m-1) \right) - f(a, b, c, n) \\
 h(a, b, c, n) &= \frac{1}{2} \left(mn(m+1) - g(c, c-b-1, a, m-1) - f(c, c-b-1, a, m-1) \right)
 \end{aligned}$$

(II) When $a \geq c$ or $b \geq c$, we fully expand the equation so that $a' = a \bmod c$ and $b' = b \bmod c$.

$$\begin{aligned}
 f(a, b, c, n) &= \sum_{i=0}^n \left\lfloor \frac{ai+b}{c} \right\rfloor \\
 &= \sum_{i=0}^n \left(\left\lfloor \frac{a'i+b'}{c} \right\rfloor + i \left\lfloor \frac{a}{c} \right\rfloor + \left\lfloor \frac{b}{c} \right\rfloor \right) \\
 &= f(a', b', c, n) + \frac{n(n+1)}{2} \left\lfloor \frac{a}{c} \right\rfloor + (n+1) \left\lfloor \frac{b}{c} \right\rfloor \\
 g(a, b, c, n) &= \sum_{i=0}^n \left\lfloor \frac{ai+b}{c} \right\rfloor^2 \\
 &= \sum_{i=0}^n \left(\left\lfloor \frac{a'i+b'}{c} \right\rfloor + i \left\lfloor \frac{a}{c} \right\rfloor + \left\lfloor \frac{b}{c} \right\rfloor \right)^2 \\
 &= \sum_{i=0}^n \left(\left\lfloor \frac{a'i+b'}{c} \right\rfloor^2 + i^2 \left\lfloor \frac{a}{c} \right\rfloor^2 + \left\lfloor \frac{b}{c} \right\rfloor^2 + 2i \left\lfloor \frac{a'i+b'}{c} \right\rfloor + 2i \left\lfloor \frac{a}{c} \right\rfloor \left\lfloor \frac{b}{c} \right\rfloor + 2 \left\lfloor \frac{a'i+b'}{c} \right\rfloor \left\lfloor \frac{b}{c} \right\rfloor \right) \\
 &= g(a', b', c, n) + \frac{n(n+1)(2n+1)}{6} \left\lfloor \frac{a}{c} \right\rfloor^2 + (n+1) \left\lfloor \frac{b}{c} \right\rfloor^2 + 2 \left(h(a', b', c, n) + \left\lfloor \frac{a}{c} \right\rfloor \sum_{i=0}^n i \left\lfloor \frac{a'i+b'}{c} \right\rfloor + \left\lfloor \frac{b}{c} \right\rfloor \sum_{i=0}^n i \right) \\
 h(a, b, c, n) &= \sum_{i=0}^n i \left\lfloor \frac{ai+b}{c} \right\rfloor \\
 &= \sum_{i=0}^n i \left(\left\lfloor \frac{a'i+b'}{c} \right\rfloor + i \left\lfloor \frac{a}{c} \right\rfloor + \left\lfloor \frac{b}{c} \right\rfloor \right) \\
 &= \sum_{i=0}^n h(a', b', c, n) + \frac{n(n+1)(2n+1)}{6} \left\lfloor \frac{a}{c} \right\rfloor + \frac{n(n+1)}{2} \left\lfloor \frac{b}{c} \right\rfloor
 \end{aligned}$$

Write the code according to the formula, in order to prevent the data that has already been counted from being double counted, we use struct to store the three functions to calculate synchronously.

```
#include<bits/stdc++.h>
using namespace std;
typedef long long ll;
const ll mod = 998244353, inv2 = 499122177, inv6 = 166374059;
int T;
ll a, b, c, n;
struct Node
{
    ll f, g, h;
};

Node solve(ll a, ll b, ll c, ll n)
{
    Node ans, tmp;
    if (!a) return (Node){(b / c) * (n + 1) % mod, (b / c) * (b / c) % mod * (n + 1) % mod, (b / c) * n % mod * (n + 1) % mod * inv2 % mod};
    if (a < c && b < c)
    {
        ll m = (a * n + b) / c;
        if (!m) return (Node){0, 0, 0};
        tmp = solve(c, c - b - 1, a, m - 1);
        m %= mod;
        ans.f = (m * n % mod - tmp.f + mod) % mod;
        ans.g = ((m * (m + 1) % mod * n % mod - 2 * tmp.h - 2 * tmp.f - ans.f) % mod + mod) % mod;
        ans.h = ((m * n % mod * (n + 1) % mod - tmp.f - tmp.g) % mod + mod) % mod * inv2 % mod;
        return ans;
    }
    tmp = solve(a % c, b % c, c, n);
    ans.f = (tmp.f + n * (n + 1) % mod * inv2 % mod * (a / c) % mod + (n + 1) * (b / c) % mod) % mod;
    ans.g = (tmp.g + (a / c) * (a / c) % mod * n % mod * (n + 1) % mod * (2 * n + 1) % mod * inv6 % mod + (n + 1) * (b / c) % mod * (b / c) % mod + 2 * (a / c) % mod * tmp.h % mod + 2 * (b / c) * tmp.f % mod + 2 * (a / c) * (b / c) % mod * n % mod * (n + 1) % mod * inv2 % mod) % mod;
    ans.h = (tmp.h + (a / c) * n % mod * (n + 1) % mod * (2 * n + 1) % mod * inv6 % mod + (b / c) * n % mod * (n + 1) % mod * inv2 % mod) % mod;
    return ans;
}

void solve()
{
    scanf("%lld%lld%lld%lld", &n, &a, &b, &c);
    Node ans = solve(a, b, c, n);
    printf("%lld %lld %lld\n", ans.f, ans.g, ans.h);
}

int main()
{
    scanf("%d", &T);
    while (T -- ) solve();
    return 0;
}
```

Then you may want to try some problems. Here are a list of ques I met before, hope you can solve them:)

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math, floor sum

Правки



	Rev.	Язык	Кто	Когда	Δ	Комментарий
☐	en29		YipChip	2025-03-26 06:18:26	2	
☐	en28		YipChip	2025-03-25 05:33:43	0	(published)
☐	en27		YipChip	2025-03-25 05:30:35	359	(saved to drafts)
☐	en26		YipChip	2025-03-25 05:06:19	2	
☐	en25		YipChip	2025-03-25 05:03:42	454	
☐	en24		YipChip	2025-03-24 18:01:07	173	
☐	en23		YipChip	2025-03-24 15:32:43	0	(published)
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☐	en20		YipChip	2025-03-24 15:28:45	6	
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